

## Deep Learning and Its Application

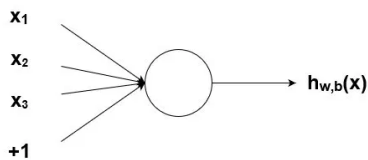
### Week 2: Practicals

1. The sigmoid function is a common activation function as follows

$$f(z) = \frac{1}{1 + \exp(-z)}$$

Given an array [-5, 5] and step size 0.1, implement this activation function in Python (refer to Section 2.1)

2. Given a network structure below



and linear formula

$$x_1 w_1 + x_2 w_2 + x_3 w_3 + b$$

Use  $x$  values calculate in Q1 as inputs and use the sigmoid function to generate outputs of nodes (refer to Sections 2.2, 2.3)

3. Let's start with the specific case of the three layer neural network below

$$\begin{aligned} h_1^{(2)} &= f(w_{11}^{(1)} x_1 + w_{12}^{(1)} x_2 + w_{13}^{(1)} x_3 + b_1^{(1)}) \\ h_2^{(2)} &= f(w_{21}^{(1)} x_1 + w_{22}^{(1)} x_2 + w_{23}^{(1)} x_3 + b_2^{(1)}) \\ h_3^{(2)} &= f(w_{31}^{(1)} x_1 + w_{32}^{(1)} x_2 + w_{33}^{(1)} x_3 + b_3^{(1)}) \\ h_{W,b}^{(2)}(x) &= h_1^{(3)} = f(w_{11}^{(2)} h_1^{(2)} + w_{12}^{(2)} h_2^{(2)} + w_{13}^{(2)} h_3^{(2)} + b_1^{(2)}) \end{aligned}$$

implement a neural network along with forward weight propagation (refer to Sections 3.1, 3.2, 3.3).

4. Understand matrix multiplication and implement the calculation (refer to Sections 3.4, 3.5).